

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Data Warehousing & Business Intelligence, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: DBMS, Big Data Analysis, Data Warehousing and Mining, Project Management, NLP, Machine Learning, Social Media Analytics

SKILLS

• **Machine Learning / Statistics:** Classification, Regression, Feature Engineering, Model Evaluation, Precision, Recall, Accuracy, Statistical Analysis, Predictive Analytics, Data Validation

• **Programming / Querying:** Python, SQL, Pandas, NumPy, PySpark, PostgreSQL, Neo4j, Scikit-learn, LightGBM

• **Visualization / Reporting:** Power BI, Tableau, DAX, Excel, VBA, Dashboard Development, Data Visualization

• **Data / Cloud / Tools:** AWS S3, EC2, Redshift, EKS, FastAPI, Docker, Git, Kafka, pgVector, GCP

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

• Designed and implemented end-to-end ML pipeline to collect, process, and analyze sensor and wearable data from 100+ users, applying signal-level feature engineering and behavioral pattern extraction to enable longitudinal predictive analytics and data-driven insights

• Developed and optimized **PostgreSQL** data models, indexing strategies, and query execution plans for high-volume datasets, improving query performance by **38%** while enabling efficient feature extraction for machine learning workflows

• Built embedding-based retrieval and summarization pipeline to enhance contextual understanding of user interactions, improving multi-turn response accuracy by 43% and supporting more reliable behavioral analysis

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

• Developed and validated NLP-based supervised ML pipelines to classify behavioral patterns from large-scale unstructured data, applying text preprocessing and feature extraction to reduce processing latency by 41% and improve model reliability for production use

• Built distributed **data pipelines** using **Kafka** and **Neo4j** Streams to enable **scalable ingestion**, transformation, and real-time analysis of high-volume unstructured datasets

• Designed production-grade inference **APIs** and deployment workflows to support reproducibility, monitoring, and consistent evaluation of machine learning outputs

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

• Built and deployed **Retrieval-Augmented Generation (RAG)** system using embeddings and **vector search**, combining data pipelines, feature engineering, and retrieval optimization to improve document retrieval efficiency by 64%

• Designed scalable **ETL** and feature pipelines using **FastAPI** and **PostgreSQL** to transform unstructured data into structured formats suitable for analytics and machine learning use cases

• Optimized indexing strategies and data processing workflows to enable low-latency querying and scalable deployment of AI-driven systems

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

• Analyzed large-scale transportation and mobility datasets using Python and PySpark, applying EDA and anomaly detection to identify operational trends and performance issues, translating findings into data-driven recommendations for business decision-making

• Built **Power BI** dashboards to track key performance indicators and improve visibility into operational performance, reducing reporting time by 40%

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

• Developed and maintained Power BI dashboards using **SQL** and **DAX**, transforming raw data into actionable insights for business decision-making across multiple domains.

• Performed data validation, cleaning, and transformation on large multi-source datasets using **SQL** and **Excel**, improving data accuracy by over 30% and reducing reporting inconsistencies, enabling faster and more reliable decision-making across sales, finance, and operations teams

Stu/Dio at Illinois

August 2025 - Present

Project Manager

• Led a **cross-functional** team of six to deliver and scale Deepcover game enhancements by leveraging user data, performance metrics, and feedback analysis to drive feature prioritization and improve games outcomes

PROJECTS

Scalable ML System for Customer Risk Prediction

• Built and validated a supervised LightGBM classification model on large-scale real-world tabular data to predict customer credit risk, applying feature engineering, cross-validation, and precision/recall evaluation to identify risk segments and drive actionable business decisions

• Performed feature engineering on behavioral patterns such as transaction frequency and spending trends, improving **model precision** by 20% while enhancing robustness and interpretability.

• Evaluated **model performance** using precision, recall, and accuracy, and designed a decision framework to translate predictions into actionable credit limit recommendations.